# PROJECT CHARTER

## BENGALURU TRAFFIC PULSE – TRAFFIC CONGESTION ANALYSIS

**Project Type:** Data Analytics Internship Project
**Domain:** Transportation and Urban Traffic Analytics
**Tools:** Python, Google Colab, Pandas, Scikit-learn, Matplotlib, Power BI and GitHub

### 1. Project Background

Traffic congestion is a major challenge in Bengaluru. Increasing vehicle movement, busy intersections, road construction and other factors can affect daily transportation.

This project analyses Bengaluru traffic data to understand congestion patterns across different areas and identify areas that may require better traffic management.

### 2. Project Objectives

- Collect and understand the Bengaluru traffic dataset.
- Clean and preprocess the dataset using Python.
- Perform data quality checks, including missing values and duplicate records.
- Analyse traffic volume, average speed, congestion and incident reports.
- Identify high-congestion and low-congestion areas.
- Apply K-means clustering to group areas based on traffic patterns.
- Develop an interactive Power BI dashboard.
- Recommend two possible traffic management interventions.

### 3. Dataset Details

**Dataset Name:** Bangalore City Traffic Dataset

**Dataset Source:**
https://www.kaggle.com/datasets/preethamgouda/banglore-city-traffic-dataset

**Dataset Size:** 8,936 rows and 16 columns.

**Date Range:** 1 January 2022 to 9 August 2024.

**Number of Areas:** 8.

### 4. Project Methodology

The project follows these stages:

1. Dataset collection and inspection.
2. Data cleaning and preprocessing.

3. Data quality checking.
4. Exploratory Data Analysis (EDA).
5. Area-wise traffic analysis.
6. K-means clustering.
7. Data visualisation.
8. Power BI dashboard development.
9. Findings and recommendations.

## 5. Tools and Technologies

- Python – Data analysis and preprocessing.
- Google Colab – Python execution environment.
- Pandas – Data cleaning and manipulation.
- Scikit-learn – K-means clustering and feature scaling.
- Matplotlib – Data visualisation.
- Power BI – Interactive dashboard.
- GitHub – Project storage and collaboration.

## 6. Expected Deliverables

- Cleaned traffic dataset.
- Data quality analysis.
- Area-wise traffic analysis table.
- K-means clustering results.
- Python analysis notebook.
- Traffic congestion charts.
- Power BI dashboard.
- Dashboard screenshots.
- Seven-slide presentation.
- README documentation.

## 7. Expected Outcomes

The project aims to identify areas with different congestion patterns and provide useful insights for traffic management.

The analysis identified Koramangala as the area with the highest average congestion (93.99), while Electronic City had the lowest (54.45).

Two possible interventions are proposed:

1. Traffic signal optimisation in high-congestion areas.
2. Better peak-hour traffic management at busy intersections.

The actual impact of these interventions would need to be evaluated through pilot testing.

## 8. Project Limitations

- The analysis depends on the available dataset.

- Only eight area-level observations were used for clustering.
- Clustering results are exploratory.
- Some congestion and capacity utilisation values reach 100.
- The analysis does not establish causal relationships.
- Intervention impacts have not been experimentally measured.

## 9. Team Responsibilities

**Member 1:**

- Dataset collection.
- Data cleaning and preprocessing.
- Data quality checks.
- Exploratory Data Analysis.
- Area-wise traffic analysis.

**Member 2:**

- K-means clustering.
- Cluster interpretation.
- Power BI dashboard development.
- Data visualisation.
- Project documentation and presentation.

## 10. Conclusion

Bengaluru Traffic Pulse uses data analytics and machine learning techniques to understand traffic congestion patterns across Bengaluru. The project provides area-wise insights, exploratory congestion clusters and an interactive dashboard to support traffic management decisions.

**Project Status:** Data analysis and Power BI dashboard completed; final documentation and submission packaging in progress.